

Thomas Verelst

PhD Researcher in Computer Vision - ESAT-PSI, KU Leuven

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Reducing inference time of deep learning by writing extra code for model compression and conditional computation.
Strong coding skills, endless curiosity and a touch of entrepreneurship.

EDUCATION

PhD candidate in Engineering Science

Computer vision and deep learning

📅 Oct 2018 - Present 📍 KU Leuven, Belgium

- PhD promotor: Prof. Tinne Tuytelaars
- Efficient convolutional neural networks (CNN), model compression and dynamic neural networks with conditional computation, reducing the computational cost of state of the art networks for classification, segmentation, human/hand pose estimation, and object detection.
- Implementation of CV algorithms for industry partners
- Teaching of courses and master theses.

Master of Science: Electrical Engineering

Embedded systems and multimedia

📅 Sept 2016 – June 2018 📍 KU Leuven, Belgium

- Magna Cum Laude (81.28%)
- with one-year exchange to EPFL Lausanne, Switzerland

Bachelor of Science

Electrical engineering - computer science

📅 Sept 2013 – June 2016 📍 KU Leuven, Belgium

- Magna Cum Laude (82.67%)

LEAD AUTHOR PUBLICATIONS

Paper details, PDFs, and other co-publications on
<https://thomasverelst.github.io/>

BlockCopy: High-Resolution Video Processing with Block-Sparse Feature Propagation and Online Policies

Thomas Verelst and Tinne Tuytelaars, *ICCV 2021 conference*

Accelerates CNNs on video by reusing features in the temporal domain.

SegBlocks: Towards Block-Based Adaptive Resolution Networks for Fast Segmentation

Thomas Verelst and Tinne Tuytelaars, *ECCV Embedded Vision Workshop 2020*

Best Short Paper Award. Reduces computational cost and memory usage of deep neural networks for semantic segmentation by adapting the processing resolution of important image regions.

Dynamic Convolutions: Exploiting Spatial Sparsity for Faster Inference

Thomas Verelst and Tinne Tuytelaars, *CVPR 2020 conference*

Reduces computational cost of CNNs by selectively applying convolutional operations using end-to-end trained attention.

Generating Superpixels using Deep Representations

Thomas Verelst, Matthew Blaschko and Maxim Berman, *CVPR 2018 DeepVision workshop*, Master thesis work (grade 17.9/20)

INTERNSHIPS AND SUMMER SCHOOLS

AI/ML R&D Internship (PhD) at Apple

Data-efficient deep learning

📅 April - August 2021 📍 Zürich, Switzerland

ICVSS 2019 summer school

Focusing on computer vision and deep learning

📅 July 2019 📍 Sicily, Italy

Internship (master) at Nokia Networks

📅 6 weeks, Summer 2017 📍 Antwerp, Belgium

Internship (bachelor) at Semicon Sp. z o.o.

📅 6 weeks, Summer 2016 📍 Warsaw, Poland

SKILLS

Languages

Dutch: native
English: fluent

French: fair
German: basic

Programming

- Proficient: Python, PyTorch, Numpy, OpenCV, MATLAB, GIT
- Familiar: C/C++, Bash, CUDA, Scikit-learn, Java
- Beginner: Tensorflow, Keras, .NET C#
- Electronics: basic VHDL, PCB design (Altium), FPGA, microcontrollers
- WebDev: HTML5, PHP, CSS3, Javascript, MySQL, jQuery, Wordpress, Flask, Bootstrap

EXTRACURRICULAR

Avid participant in tech events such as hackathons and workshops. When not programming, I am probably hiking, photographing or skiing in the Alps.

- Hackathons: building prototypes in 48 hours
 - HackJunction2020 Connected: challenge winner - smart shopping assistant app
 - HackZurich 2018 (Europe's largest hackathon): finalist and IBM challenge winner - risk forecasting app
 - HackTheAlps2018: public award and challenge winner - ski area estimation from aerial imaging
- AFC Participant and Student Consultant (2015-2017): IT consulting in a multi-disciplinary team of 5. Leadership, presentation & productivity workshop tracks.
- Full-stack web developer (2012-...): development and support of a templating framework. PageRank 5 with 500.000 visitors since 2012. <http://metro-webdesign.info/>

REFERENCES

PhD promotor: Prof. Tinne Tuytelaars
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